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Third places, the connective fibre of cities and high-tech entrepreneurship

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ABSTRACT

Urban 'third places' foster informal interactions and face-to-face contact that is critical to the development of new innovations and start-up businesses. Our research utilises extensive microdata in combination with telephone interviews with over 200 start-up companies in Stockholm, Sweden. Our findings indicate that access to third places has a significant impact on the number of new high-tech start-ups (both by entrepreneur place of residence and work) in their vicinity. This suggests that third places play a demonstrable role in innovation and economic growth.

KEYWORDS

third places; entrepreneurship; innovation; high-tech start-ups; start-up ecosystems

JEL L26, R12, R23

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1. INTRODUCTION

Cities are typically perceived as serving two roles. On one hand, they are viewed as centres of economic activity, that is, as places for employment, innovation and productivity (Jacobs, 1969; Lucas, 1988). This stems from their increased density and diversity, their clusters of firms, their human capital, the economic assets they bring together, and their ability to create robust innovation and start-up ecosystems. On the other hand, cities are also centres of consumption (Glaeser et al., 2001) that provide access to amenities such as bars, restaurants, gallery spaces, cafés, nightclubs, theatres, concert halls and fitness/health centres such as yoga, gyms and dance studios (Clark et al., 2002; Lloyd & Clark, 2001; Silver et al., 2010, 2011).

These two dimensions of city life come together through a unique set of organisations that sociologists have dubbed 'third places' – places where people come together that are neither home nor work but blend elements of both (Oldenburg, 1989; Oldenburg & Brissett, 1982). Third places work to connect the productivity and consumption sides of cities, enabling people not only to enjoy each other's company but also to share information and knowledge and to build networks and business relationships. As such they serve as critical components of the connective fibre crucial to the innovative capacities of cities. Saxenian (1994) highlighted the importance of such spaces to the face-to-face social interactions that enable knowledge spillovers. She identified the role played by the Wagon Wheel, a bar in Silicon Valley, where engineers and entrepreneurs from different companies would connect after working hours, dubbing it the 'Fountainhead of the Semiconductor Industry'. Recent research finds that accessibility to bars, restaurants, hotels and coffee shops facilitates social interactions that generate knowledge spillovers and promote innovation (Roche, 2020). Furthermore, urban economists and economic geographers have long noted a connection between urbanisation economies and entrepreneurship (Acs et al., 2011; Audretsch et al., 2015; Tavassoli et al., 2021). However, when it comes to the mechanisms of the relationship, research has often referred to the black box of knowledge spillovers. A stream of studies has sought to unpack the various components or dimensions of innovation (Roche, 2020), or start-up ecosystems (Kenney & Florida, 2000; Saxenian, 1994). While the role of third places in such contexts has been suggested in the qualitative literature, additional empirical research is needed to identify more precisely the role played by such third places in entrepreneurship and innovation dynamics (Fang et al., 2022).

Our research contributes to this literature by analysing more closely the role of third places – particularly bars, restaurants and cafés – in promoting the formation of entrepreneurial high-tech start-ups in cities. The central

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question we pose is to what degree does access to third places increase the likelihood of high-tech start-ups. The intuition behind this question is twofold: (1) if people live or work close to places with higher accessibility to third places, they should be more likely to meet other people and exchange knowledge and ideas. The importance of face-to-face interaction for knowledge transfer – even in highly digitalised environments and industries – has been demonstrated in the literature for some time (Battiston et al., 2020; Cowgill et al., 2009; Roche et al., 2022). However, we are also interested in (2) the less examined role that third places play in the actual planning of the start-up itself, as coffee shops, bars and restaurants can function as a kind of 'quasi'-office for early stage start-ups. In either case, the link between third places and start-ups is particularly relevant today as cities strategise for downtown recovery stemming from the impact of the COVID-19 pandemic, and the shift to remote work, which have left many downtowns with less activity than before the pandemic.

To examine more closely the role of third places in start-up formation, our research focuses on the municipality of Stockholm, which provides a useful case study for this question as it ranks as one of the world's leading start-up ecosystems. The Stockholm labour market is home to approximately one-fourth of the country's population, but approximately 35% of all engineers, 46% of all computer scientists (Mellander & Florida, 2007) and 76% of all venture capital investments, 90% of which went to firms located in the city centre (Adler et al., 2019). Stockholm is home to many successful start-ups, such as Spotify, Skype, Minecraft and Candy Crush. It is also home to great universities, such as Stockholm University, Karolinska Institute, The Royal Institute of Technology and Stockholm School of Economics. It is a centre for urban amenities, teaming with third places of wide appeal.

Our analysis includes a mix of microdata and phone survey data to examine the connection between third places and start-ups. We first map everyone who started a high-tech firm in Stockholm between 2010 and 2019. We identify both their place of residence and their place of work (within 250 m^2 grid cells) the year before the actual launch of the new firm. We then analyse the relationship between spatial access to third places and the number of start-ups while controlling for several grid characteristics which can relate to entrepreneurship. All estimations include neighbourhood fixed effects to account for the heterogeneity of neighbourhoods in Stockholm. In addition, we conducted a telephone survey of over 200 entrepreneurs in Stockholm in October 2020 to better understand their perceptions regarding the role of third places in fostering their business ideas and start-ups.

Our key findings confirm that third places play a role in high-tech business formation. Indeed, access (proximity) to third places is significantly associated with new business creation across all model specifications, with a slightly larger impact based on the entrepreneur's place of residence than place of work (although both are significant). At the same time, 24% of surveyed entrepreneurs indicated that they even planned the formation of their firm primarily in third places such as restaurants and bars, compared with 25% at home and 40% at the office. In addition, 20% of entrepreneurs indicated in an open-ended response that third places were 'very important' to the early stages of the start-up to meet and discuss issues related to the business, compared with 21% in the office and 15% online. Taken together, these findings provide suggestive evidence of the importance of third places in entrepreneurship, particularly for planning in the early stages of firm formation. We believe our results can be generalised beyond the context of Stockholm to other large cities and urbanised areas with high densities of third places.

The remainder of this paper is organised as follows. The next section provides a deeper overview of the relevant theoretical framework. We then turn to our data, identification strategy and empirical methods. We include a section where we descriptively discuss the location of third places as well as start-ups in Stockholm. Following that, we present and analyse the key results from the regression analyses and the phone survey we conducted. The last section discusses these results and their salience relative to prior research and ongoing studies about cities, amenities and innovation.

2. THIRD PLACES, KNOWLEDGE-SHARING AND ENTREPRENEURSHIP

Our research probes the role of third places not just as spaces of consumption but in spurring the knowledge sharing that motivates innovation, and entrepreneurship. Innovation and entrepreneurship in cities are usually seen as the products of density, diversity, and the hard economic or technological assets of cities such as high-tech firms, highly educated human capital, university research and venture capitalists. More recent research points to the role of face-to-face interaction and so-called 'buzz' in accelerating the flow of knowledge exchange and information in cities (Asheim et al., 2007; Currid & Williams, 2010; Storper & Venables, 2004).

Research on entrepreneurship dates back to Schumpeter (1934) whose work very much focused on the entrepreneur and their abilities to perceive new technological and economic emerging out of innovation. A huge literature builds on these basic insights on the role of innovation and entrepreneurship in driving economic development (Aghion & Howitt, 1990; Griliches, 1957; Grossman & Helpman, 1991; Nelson & Winter, 1982).

Jacobs (1969) long ago argued that innovation and economic development are processes that occur just as products of innovative large and small firms. Cities function to attract a diverse set of actors, talented and skilled people who drive economic development. Lucas (1988) formalised Jacobs' ideas with the construct of 'human capital externalities' which serves as the fundamental mechanism of innovation and economic development. The connection between human capital and economic growth has been widely documented in research, at both the national (Barro, 1991) and regional levels (Berry & Glaeser,

2005; Florida, 2002b; Glaeser, 2000; Rauch, 1993). Other research has documented the link between why skilled cities grow faster than low-skilled cities due to higher levels of productivity (Glaeser & Saiz, 2003; Rauch, 1993).

Urban economists and geographers have also long identified that such innovation and entrepreneurship is not just the province of firms per se but occurs in geographical clusters (Acs et al., 2002; Anselin et al., 1997; Jaffe et al., 1993). A large and growing body of studies identifies the mechanism driving innovation and entrepreneurship in such clusters to be knowledge spillovers (Anselin et al., 1997; Audretsch & Feldman, 2004; Jaffe et al., 1993). Related research documents the geographical proximity for interactive learning and innovation (Balland et al., 2015; Boschma, 2005; Fritsch & Kublina, 2018). Storper and Venables (2004) break down face-to-face interaction into four main features that create 'buzz' in a place: (1) it is an efficient technology for communication; (2) it helps solve incentive problems; (3) it facilitates both learning and socialisation; and (4) it provides psychological motivation. Buzz in cities makes people more productive and creates more cross-fertilisation between different sector networks, such as science and tech, creative and cultural industries, finance and business, and/or the power and influence sectors. Guiso and Schivardi (2011) find that knowledge spillovers happen more efficiently in areas with many entrepreneurs and thus the probability that individuals develop entrepreneurial skills is higher. Andersson and Larsson (2014) show that the presence of entrepreneurs in a neighbourhood influences the emergence of new entrepreneurs in the area, arguing that it is due to social interactions among people in the neighbourhood. Sorenson and Audia (2000) suggest that if an area is highly entrepreneurial, residents might be more motivated and have higher self-confidence that they can also make it if they start their firm. Minniti (2005) argues that social interactions between established entrepreneurs and people who want to start their firms lower the uncertainties about the start-up process. These studies highlight the role of social interactions in motivating the knowledge spillovers which underpin innovation and start-up formation.

A growing body of research focuses on the consumption side of cities in the clustering of talent and in spurring increases in innovative and creative output. Glaeser and Saiz (2003) show that high-skill workers tend to cluster around consumption opportunities. Kolenda and Liu (2012) find that advanced, fast-growing creative industries and their employees are more likely to locate in city cores. Other research examines the role of amenities and 'scenes' in creative output (Clark, 2003). Silver et al. (2011) suggest that scenes and technology industry clusters are contextually joined together.

Our research builds on these contributions to focus on the role of third spaces such as bars, cafés, restaurants, galleries and music venues in fostering the knowledge spillovers that shape innovation entrepreneurship (Jacobs, 1961; Oldenburg, 1989). Their role is especially important to interaction and knowledge sharing in the creative industries (Currid, 2007; Storper & Venables, 2004). Fang et al. (2022) find that proximity to coffee shops is significantly related to start-up formation. Furthermore, third places play a critical role in bridging the production and consumption functions of cities. Most research in urban economics and urban geography sees innovation and start-ups as outcomes of the production side of cities. Third places are typically bundled with a set of amenities that form the consumption side of cities. But they also play a crucial role in mobilising face-to-face communication and social interaction which is central to knowledge spillovers (Landry, 2000). As their very name implies, third places are arenas that are neither work nor home per se, but support elements of both. We thus hypothesise that they play a key role in fostering social interaction and knowledge-sharing, which motivate innovation and start-up formation. They are a critical element of the human connectivity that shapes innovation, entrepreneurship and economic development in cities.

3. DATA, IDENTIFICATION AND METHOD

To get at these issues and questions, we use register geocoded data from Statistics Sweden. We first identify those who started a new high-tech firm and the exact location of their workplace and their place of residence within 250 m^2 grid cells during the period 2010–19. We can also calculate their geographical proximity to cafés and restaurants as a proxy of all types of third places in the municipality of Stockholm the year prior to the start-up. Our unit of analysis is the 250 m^2 grid squares, which addresses the very localised nature of knowledge spillovers through social interactions (Arzaghi & Henderson, 2008). The dependent variable is therefore the number of firms started by individuals working and/or residing in each 250 m^2 grid square. Following the prior literature, we expect that accessibility to meeting places increases the face-to-face interactions between individuals working (living) in the same square.

Our independent variable of interest, *Third place Access*, is created by calculating a spatial access score for each grid cell using the Access package (v.1.1.3) in PySAL (Saxon et al., 2021). The idea of the access score is to provide a more realistic measure of weighted spatial accessibility to a given type of amenity – based on the fact that accessibility decays with distance rather than being a hard cut-off at the arbitrary grid cell boundaries (Isard, 1960). Moreover, empirical literature has also shown that social interactions are very local (Andersson & Larsson, 2014; Arzaghi & Henderson, 2008). We therefore calculate the Euclidean distance from each grid cell i centroid to the 10 nearest grid cell centroids j that contain third places, discounted by a distance-decay factor (negative exponent) of 0.25. Each decay-discounted ij link is multiplied by the number of associated third places at j; these weighted distances to nearby third places are then summed by grid cell origin (i) to provide an aggregate measure of high accessibility *to* nearby third places *from* each grid cell. In this way, we estimate how the number of new firms in time $t+1$ depend on

accessibility to third places in time t. Since all squares are of equal size, this variable becomes a density measure.

Empirically, we should be aware of the issue of omitted variable bias. Economic activity and entrepreneurship might be simultaneously determined in specific areas of the city, for instance, in the city centre. These areas, in turn, may be also attracting people who are more entrepreneurial to begin with. There is a large literature, starting with Florida (2002b), arguing that more creative individuals are also more likely to be in areas characterised by more amenities. Following Roche (2020), we attempt to address this issue by including neighbourhood fixed effects. This way we can hold the unobservable characteristics of a neighbourhood such as culture or access to different amenities constant. Neighbourhoods in our case are regional statistical areas (RegSO). Statistics Sweden introduced RegSO as a definition of neighbourhoods designed specifically to facilitate the statistical analyses of socio-economic segregation. RegSO is a good geographical unit for analysing neighbourhoods because it accounts for demographic conditions based on specific geographical borders, such as streets, waterways and railways, representing real neighbourhoods rather than arbitrary administrative areas.[1] The identification comes therefore from within the neighbourhoods. We test the following specification where our unit of observation is a 250 m^2 grid square:

$$Entrepreneurship_{it+1} = \varphi_0 + \varphi_1 Third\ place\ Access_{it} + \Theta X_{it} + \mu D_n + \gamma D_t + \varepsilon_{it}$$

where *Entrepreneurship* is the number of high-tech firms started in time $t + 1$ by people who worked (lived) in square i in time t. We focus specifically on high-tech firms, which we assume are dependent to a higher degree on knowledge spillovers. $Third\ place\ Access_{it}$ denotes the accessibility measure to third places, here defined as cafés, bars and restaurants from the centre of the square.[2] We allow for a one-year lag between the two, due to the time it takes to plan and set up a business. Θ is a vector of control variables, which are presented in Table 1 together with their mean and standard deviation (SD).[3] The values in parentheses denote descriptive statistics for the square of residence. All of them are also measured in time t. D_n and D_t denote respectively the neighbourhood and year fixed effects. ε_{it} is the traditional error term. Again, we repeat that all squares are of equal size, which implies that all count variables become measures of density.

The second identification problem is the one of selection, where high-tech firms possibly select themselves in the city centre where the bars are also located. This way neither the location of bars, nor the one of entrepreneurs is random. To take care of this issue we use an instrumental variable (IV) approach, following other studies of similar set-ups (Roche, 2020). The idea with an instrument is that it should be correlated with where third places are today, but not directly affect the outcome variable, that is, entrepreneurship other than the effect from the bars and restaurants (Angrist & Pischke, 2008). The instrument we use is accessibility to where the cultural workers lived in the city of Stockholm in 1882. We rely on

Table 1. Variables used in the empirical model.

Variables	Measured as …	Mean (SD) work	Mean (SD) live
Entrepreneurship[a]	Number of high-tech firms started by people who worked (lived) in square in $t + 1$	1.923 (7.196)	2.291 (4.139)
Third Places	Accessibility to third places	123.7 (329.9)	125.7 (337.9)
Education	Share of workers (residents) with a university degree	0.278 (0.198)	0.333 (0.16)
Employment	Employment (population) size	253.7 (729.7)	313.5 (368.4)
Creative Occupations	Share of workers (residents) in a creative occupation	0.286 (0.251)	0.346 (0.225)
Average Age	Average age in the square for employees (residents)	44.52 (5.888)	47.67 (6.588)
Income	Mean labour and capital income in the square (SEK100)	2845 (2246)	3438 (24,939)
University	Accessibility to university campuses[b]	3.308 (15.85)	3.285 (15.70)
Share Female	Share of female workers (residents) in the square	0.44 (0.219)	0.495 (0.099)

Note: In the regressions, entrepreneurship, third places, employment, income and accessibility to universities are included in the ln form.
[a]For the distribution of the high-tech entrepreneurship, see Figure A1 in Appendix A in the supplemental data online.
[b]Stockholm has three universities: Stockholm University, the Royal Institute of Technology and Stockholm School of Economics, which consist of different campuses in the city.

historical data of the Stockholm Address Calendar which are digitalised by Stockholm City and available online. We manually extracted each individual from the register book into the dataset.

These calendar files, which functioned as a 'telephone directory of the time', contain detailed information about the inhabitants of Stockholm including the address where they lived and their occupation. While it was not compulsory for the individuals to register in the address calendar, we have about 16,500 individuals in the dataset, which corresponds to around 10–15% of the whole population of Stockholm at the time.

To define the cultural workers, we use the occupations mentioned in the codebooks. In many cases, the occupation is no longer available today, and in these cases, we have tried to match it to the occupation closest to today's labour market. For example, a horse driver has been transferred into being a taxi driver. We define cultural workers as people working in a cultural ('bohemian') occupation, such as a musician, singer or dancer. Assuming that there is some path dependency in places which are considered as 'creative', we argue that the location of the cultural workers in 1882 should be correlated with where the third places are located today. Figure 1 also shows the parts of Stockholm in which they were living. It is worth noting that the city of Stockholm was much smaller in the late 1800s. Therefore, the geographical extent of the city then would be considered the inner city of the region today – all of which are now highly expensive neighbourhoods with many third places. However, according to an analysis of distribution of population by occupation, the cultural workers were less segregated than other groups of the society such as individuals working for the royal court, nobility, fishing and farming occupations, and university graduates. Moreover, a vast body of literature discusses the correlation between creative, third places and bohemian culture, suggesting that they thrive in similar environments – both historically and today (Currid-Halkett, 2009; Florida, 2002a; Florida & Mellander, 2009; Markusen & Schrock, 2006; Seigel, 1999).

While the inclusion of fixed effects combined with the use of IVs helps with identification issues, claiming causal effects is always cumbersome in the absence of exogenous variation in the identification strategy. For this reason, we further completed the analysis with a conducted phone survey. We randomly selected 1073 high-tech firms in Stockholm that started in 2016 and through a phone survey examined the importance to them of third places in

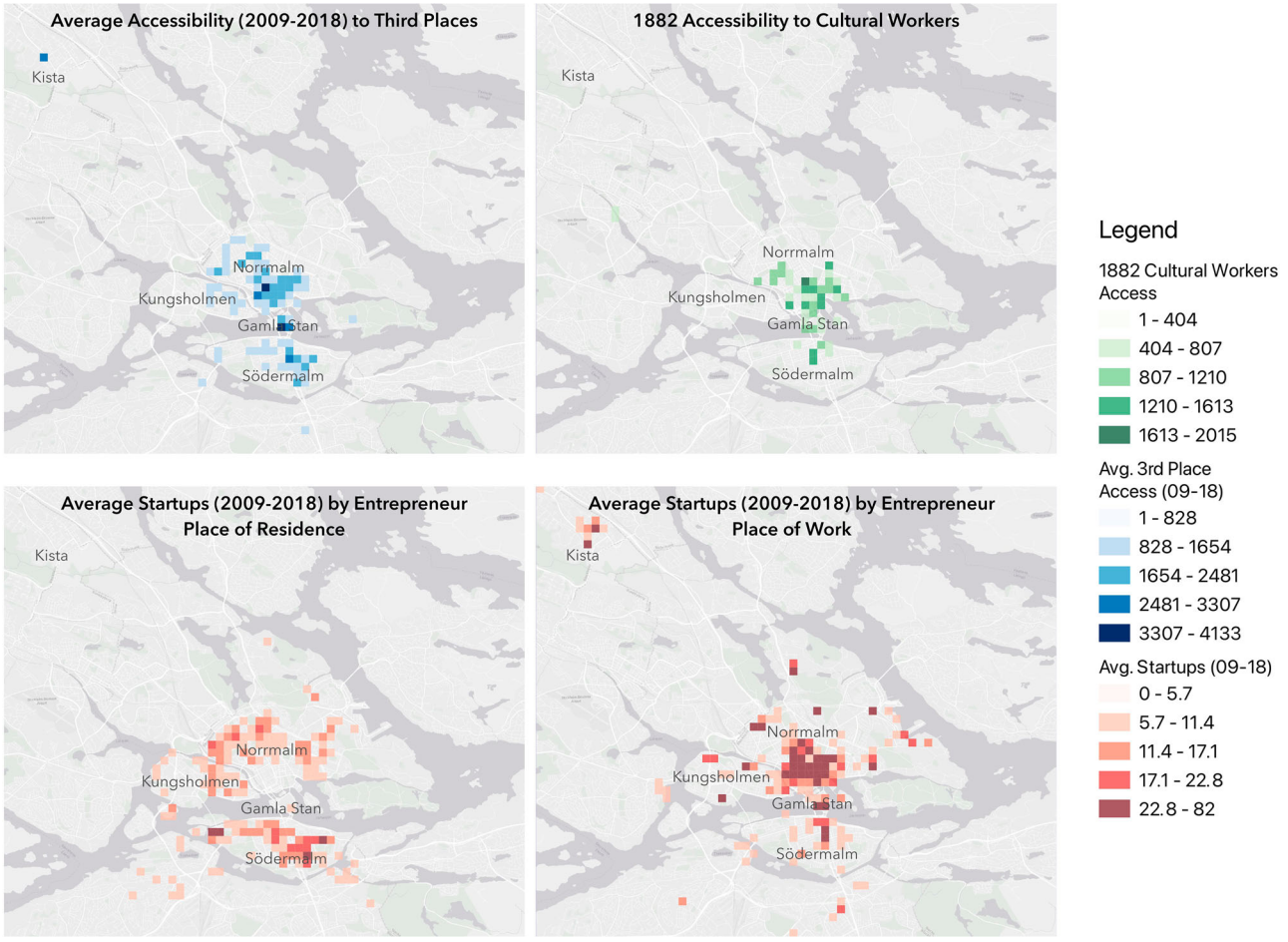


Figure 1. Spatial distribution of average yearly (2009–18) third place access scores, accessibility to the 1882 proportion of cultural workers employment, and average yearly (2009–18) number of entrepreneurs (by workplace and residence) across Stockholm.

their start-up phase. In total, 228 firms responded, a response rate of just above 21%. We obtained information regarding the size of the firms that responded as well as those that did not. The difference is, however, statistically insignificant, so we do not have a selection on firm size. Given that there are no large differences, at least on the variables we can observe, we believe that the answers obtained are representative of high-tech start-ups in general. In the survey, we asked various general questions regarding the firm, where their idea first came up, whether it involved conversations with other people, as well as their opinions about the importance of meeting places for them during the start of the firm. Through an open question, we also allowed respondents to express in their own words why they believed the meeting places to be important.

4. ACCESSIBILITY TO THIRD PLACES

Figure 1 shows the spatial distribution of third place access and entrepreneur locations across the Stockholm municipality. The top left panel shows the distribution of average yearly (2009–18) third place access scores (the independent variable of interest). The top right panel shows accessibility to the 1882 proportion of cultural class employment (the IV), while the remaining two panels show the average yearly (2009–18) number of high-tech[4] entrepreneurs by workplace location (bottom right) and residence location (bottom left), the dependent variables of interest.

In general, the highest concentration of third places is in the central parts of the city, including Gamla Stan, Södermalm, Norrmalm, Östermalm and, to a slightly lesser extent, Kungsholmen. These localities make up the inner city of Stockholm and are parts of the municipality that to a large extent compose the historical area of the region. Today they are home to an extensive number of bars, cafés, nightlife venues and retail shopping stores. The top two panels of Figure 1 indicate that they are also the areas most favoured by entrepreneurs. Interestingly, entrepreneur workplace locations tend to be concentrated in the very centre, for example, Östermalm and Gamla Stan, while residence locations tend to be on the 'periphery' of that centre, in places such as Kungsholmen and Södermalm. However, the pattern of third place access is not descriptively much different between these two locations.

Beyond the centre, we find a secondary concentration of third places and entrepreneur workplace locations in Kista, in northern Stockholm. While slightly further away from the city centre, this pattern is not surprising, since in recent years Kista has become one of the largest information and communication technology (ICT) clusters in Sweden, informally referred to the Silicon Valley of Stockholm. High-tech firms such as Ericsson, IBM, Samsung and Nokia are located there, which have attracted a lot of high-skilled workers into this area.

While Figure 1 suggests a clear spatial relationship between third place access and the residence and workplace locations of entrepreneurs, a more detailed empirical analysis is required to see whether this relationship holds while controlling for other related factors (including density and employment in creative occupations).

5. RESULTS AND ANALYSIS

As previously discussed, to establish the importance of third places for high-tech entrepreneurship, we combine quantitative empirical models together with a qualitative survey of Swedish high-tech entrepreneurs. We look first at the quantitative data.

5.1. Accessibility to third places and start-ups

Table 2 shows the effects estimations for the start-ups in high-tech industries from the 250 m^2 grid cells where people work (columns 1–3) and where they live (columns 4–6). In columns (1) and (3) neighbourhood fixed effects are not included.

The accessibility to third places estimates shows a positive and statistically significant relationship in all the estimates. This provides suggestive evidence for the importance of third places in the start-up process of high-tech firms. These results indicate that access to third places is positively related to the number of high-tech start-ups. The positive and significant relationship holds when looking both at the place of work and the place of residence of the entrepreneur. The variable is significant also when we control for employment density, creative workers in the square, age structures, income, gender composition in the location and access to universities. In effect, accessibility to third places increases the number of newly started high-tech establishments from people who live or work in the vicinity. When looking at the size of the coefficients, the estimations in Table 2 show that a 1% increase in access to third places increases high-tech entrepreneurship by about 0.04% in both estimations. To help with the interpretation of the magnitude, it is easier to discuss in terms of standard deviation change rather than percentage change. For the work model, a 1 SD increase in accessibility to third places is associated with a 0.09 SD increase in entrepreneurship. This is a similar magnitude for both estimations.

There may be a few mechanisms at work behind this observed relationship, though they are difficult to pinpoint from the quantitative analysis. First, it might be that third places provide an increased opportunity for face-to-face interactions and exchanging thoughts and ideas. This would be in line with the well-developed theory of agglomeration economies previously discussed in this paper. When individuals are close to each other, the possibility of exchanging knowledge and information becomes easier. Duranton and Puga (2004) consider this as learning, where they see those processes aiding the generation and diffusion of knowledge accumulation. Learning is often associated with such face-to-face interactions, created through the flow of knowledge across skilled individuals (Lucas, 1988; Jovanovic & Rob, 1989). We, therefore, analyse the positive and significant result as suggestive

Table 2. Ordinary least squares (OLS) and fixed effects (FE) estimations on the number of high-tech start-ups in Stockholm, 2010–19.

	Work		Residence	
Variables	**OLS (1)**	**FE (2)**	**OLS (3)**	**FE (4)**
Third place Access	0.082***	0.031***	0.093***	0.027***
	(0.009)	(0.005)	(0.008)	(0.005)
Education	0.524***	0.189***	1.158***	0.371***
	(0.064)	(0.036)	(0.108)	(0.094)
Density	0.304***	0.258***	0.374***	0.329***
	(0.017)	(0.018)	(0.018)	(0.016)
Creative Occupations	0.521***	0.585***	0.355***	0.401***
	(0.037)	(0.061)	(0.042)	(0.075)
Average Age	0.003*	−0.000	0.003	−0.001
	(0.002)	(0.001)	(0.002)	(0.002)
Share Female	−0.341***	−0.289***	−0.562***	−0.389**
	(0.044)	(0.041)	(0.208)	(0.163)
Income	−0.026**	−0.064***	0.046	0.012
	(0.011)	(0.011)	(0.029)	(0.024)
University Access	0.118***	0.083***	0.041***	−0.025*
	(0.015)	(0.015)	(0.013)	(0.014)
Neighbourhood FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Observations	21,642	21,642	21,104	21,104
R^2	0.643	0.731	0.556	0.645

Note: OLS and FE estimations for the number of high-tech start-ups in $t + 1$ from the individual working (living) there in time t. Coefficient for the constant is not reported. Standard errors clustered at the neighbourhood level are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

evidence for the importance of face-to-face interactions for the spillover of knowledge.

A second mechanism for our finding may be that the exchange of knowledge is facilitated between individuals working at different companies, given the greater accessibility of neutral places to meet and discuss opportunities for future collaborations. For instance, previous work by, for example, Vinodrai (2006), showed how downtown Toronto functions as a meeting arena for creative workers such as designers from different companies, who profit from the opportunity to meet across employers at bars and restaurants. This mechanism would also relate to Saxenian (1994), who discussed the importance of such places for engineers to meet and exchange ideas after working hours. In addition, we discuss a potential third (currently under-theorised) mechanism that emerged from our survey results in section 5.3: that third places provide a physical space for the formal work of planning and executing a start-up.

Besides those main estimates, results for our control variables show that working in denser squares, in squares with more creative occupations as well as a higher educated workforce increases the probability of people starting a high-tech firm a year later. On the other hand, the presence of more females is negatively correlated with the entrepreneurial outcomes. This is not a surprising finding, since much literature has previously argued that women are underrepresented in the high-tech sector (Mayer, 2008; Stanworth, 2000). When looking at columns (3) and (4), where the results are based on place of residency, the results show that living in denser squares, and blocks with more educated and creative individuals is positively associated with high-tech entrepreneurship. Just like in the previous case, the share of females is negatively correlated to high-tech start-ups.

While the RegSO neighbourhoods are small and should capture the differences across different neighbourhoods of Stockholm, the results in Table 2 do not control for how affordable it is to live in different parts of the city, which might affect the location of income-constrained individuals. We therefore re-estimate the original models including the share of individuals living in a rental apartment in each square (Table 3). Unfortunately this variable is only available after 2013, thus providing a smaller number of observations. The results remain largely unchanged.

To further ensure the stability of the results, we also run a few alternative versions of the analysis (all results are available in Tables A1–A8 in Appendix A in the supplemental data online). First, we change the dependent variable to all entrepreneurs instead of high-tech entrepreneurs (see Table A1 online). We also run an analysis with a binary dependent variable that takes the value of 1 if at

Table 3. Ordinary least squares (OLS) and fixed effects (FE) estimations on the number of high-tech start-ups in Stockholm, including the share of rentals, 2010–19.

	Work		Residence	
Variables	**OLS (1)**	**FE (2)**	**OLS (3)**	**FE (4)**
Accessibility to Third Places	0.080***	0.028***	0.085***	0.025***
	(0.008)	(0.004)	(0.008)	(0.005)
Share of Rentals	−0.172***	−0.106***	0.047	−0.122***
	(0.031)	(0.028)	(0.038)	(0.036)
Education	0.493***	0.179***	1.238***	0.383***
	(0.061)	(0.035)	(0.121)	(0.112)
Density	0.316***	0.249***	0.380***	0.340***
	(0.019)	(0.020)	(0.019)	(0.017)
Creative Occupations	0.472***	0.587***	0.268***	0.256***
	(0.044)	(0.064)	(0.039)	(0.085)
Average Age	0.004**	0.000	0.003	−0.003
	(0.002)	(0.001)	(0.002)	(0.002)
Share Female	−0.363***	−0.248***	−0.032	0.058
	(0.048)	(0.037)	(0.273)	(0.196)
Income	−0.019	−0.053***	0.088**	0.012
	(0.013)	(0.012)	(0.038)	(0.031)
University Accessibility	0.143***	0.069***	0.069***	−0.020
	(0.021)	(0.023)	(0.016)	(0.014)
Neighbourhood FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Observations	13,790	13,790	20,616	20,616
R^2	0.647	0.751	0.484	0.531

Note: OLS and FE estimations for the number of high-tech start-ups in $t + 1$ from the individual working (living) there in time t. Coefficient for the constant is not reported. Standard errors clustered at the neighbourhood level are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

least one individual starts a high-tech firm in a given 250 m^2 grid cells (both based on place of establishment and residence) (see Table A2 online). Note that in this case the accessibility to third places is statistically insignificant in the fixed effects estimation of the residence model. However, this is one of the few cases where we see such a result in the robustness checks, indicating that our main conclusions are largely robust to the choice of specification. We also show the results when we use a zero-inflated negative binomial model rather than an ordinary least squares (OLS) to account for the many squares with zero entrepreneurship, a pattern shown in Figure A1 in Appendix A online. These results are shown in Table A3 online.

We also want to ensure that the findings are not driven by the way we construct the accessibility measure. Table A4 online shows the results when the log number of third places in the square is used instead of the accessibility. Table A5 online shows the results when we separate the different types of third places into two different categories: hotels and youth hostels (NACE codes 55101–55201) and restaurants, canteens and bars (NACE codes 56100–56300). Note that we can unfortunately not separate between restaurants and bars because due to Swedish law all bars must serve food. Therefore, they are all registered as restaurants. Interestingly, while the results are unchanged for the workplace model, hotels are statistically insignificant for the residence model, indicating that bars and restaurants are more important. To take into account the issue of spatial concentration of entrepreneurs within each grid cell and the relative rarity of the start-up event, we test a specification in Table A6 online that uses a spatial empirical Bayes (SEB) estimator to smooth the dependent variable. The basic concept of the SEB smoother is that it shrinks the rate observed in a given grid cell towards the mean rate of the surrounding grid cells (in this case, the eight nearest neighbours) based on the variance of the rate in this spatial neighbourhood (Anselin et al., 2004; Clayton & Kaldor, 1987; Credit, 2019). The resulting estimate can be thought of as the overall 'expected probability' of the entrepreneurship rate (per m^2) in each grid cell. Results are robust to all these specifications. Table A7 online shows the results when we include a variable that captures the ratio between the number of employees above the age of 16 divided by the number of individuals above the age of 16 living in each square. That gives a proxy for whether the square is mainly characterised by people living there or people working

there. While the results are robust and stable, it is however worth pointing out that Stockholm is quite a mixed city in terms of where people live and whether firms are located. There is not a clear central business district with only companies and no people living around there. Last, Table A8 online shows the results where we have excluded the top 95th and 99th percentiles to ensure that it is not a few squares at the top of the distribution that are driving all the results.

As discussed above, it is still difficult to establish whether the third places affect or cause entrepreneurship or if we are dealing with a selection issue. It could be the case that people working in places characterised by higher accessibility to bars and restaurants would start high-tech firms anyway. Thus, the relationship between third places and entrepreneurship would be spurious and driven by simply working or living in the city centre rather than specifically having high accessibility to places which facilitate face-to-face interactions. For this reason, below we combine the regression results with an IV approach as well as relying on a phone survey with high-tech entrepreneurs from Stockholm.

5.2. Instrumental variable (IV) estimations

To take care of this issue with selection and that high-tech firms possibly select themselves in the city centres where also the bars are located, we use an IV approach which was described in more detail above. We re-run the analysis in Table 2, but now also include our instrument (access to bohemian, cultural workers in Stockholm in 1882 to instrument the existence of bars and restaurants today). The accessibility score is calculated in the same way as for the bars and restaurants.

However, there is one caveat with using historical variables such as this: Stockholm was smaller in 1882 than today. Therefore, the areas outside the centre of the city (today) will show up with very low accessibility to 1882 cultural workers simply because the size of the city itself was different, which might then affect the IV. For this reason, we run the IV estimations only on those neighbourhoods located in the centre of the city (i.e., those within Stockholm in 1882): Gamla Stan, Kungsholmen, Långholmen, Norrmalm, Riddarholmen, Skeppsholmen, Södermalm, Vasastaden, Östermalm and Djurgården. The results are shown in Table 4. However, in Table A9 in Appendix A in the supplemental data online we also show the results with all the neighbourhoods for the municipality of Stockholm, which are consistent with the main results of the paper.

The third place access variable remains positive and significant when we include the instrument, both with and without control variables. This suggests that there is a significant relationship between access to third places and high-tech start-ups. Note, however, that the magnitude of the coefficients for the IV are larger than those of the OLS, especially in the first two columns. While there can be several reasons for the observed difference in coefficient size (e.g., Card, 2001), it could indicate that the behaviour of a subpopulation is being shifted due to the instrument as the IV and OLS measure slightly different effects. However, the value of the *F*-statistics is reassuring in all the columns given that it is above the usual threshold of 10, indicating that we do not have a problem with weak instruments. The second-stage results for the control variables is shown in Table A10 in Appendix A in the supplemental data online.

5.3. Survey results

The results of the quantitative analysis suggest that access to third places is important in the start-up phase for many high-tech firms. We supplement these findings with the results from a telephone survey conducted in October

Table 4. Instrumental variable (IV) estimations on the number of high-tech start-ups in Stockholm, 2010–19.

	Work		Residence	
Variables	**(1)**	**(2)**	**(3)**	**(4)**
Second stage				
Third place Access	0.427***	0.545***	0.066***	0.137**
	(0.034)	(0.132)	(0.014)	(0.055)
First stage				
Cultural Workers 1882	0.218***	0.081***	0.379***	0.131***
	(0.012)	(0.015)	(0.015)	(0.018)
Control variables	Yes	Yes	Yes	Yes
Neighbourhood FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
First-stage *F*-statistics	323.6	27.97	636	52.79
Observations	3297	3297	2564	2564

Note: IV estimations for the number of high-tech start-ups in $t + 1$ from the individual working (living) there in time t. The location of third places is instrumented through the location of cultural workers in Stockholm in 1882. Coefficient for the constant is not reported. Standard errors are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

2020. The interviews were with the actual entrepreneur behind the high-tech start-up, and we asked questions about the role of different kinds of places during different phases of the start-up process. We received 228 responses, a response rate of 21%.

Table 5 shows the descriptive statistics of the firms that answered the survey and the whole population of high-tech start-ups in Stockholm in 2016. It shows that we have a slight over-representation of highly educated individuals among those who answered our survey. We can also see in the survey data that the average size of the firm is slightly larger. Nevertheless, these numbers are in general similar, which indicates that the results from the survey should be generalisable for all start-ups in the area. Interestingly, 42% of those who answered the survey are serial entrepreneurs who have started more than one firm (before or after 2016), compared with 39% in the register data.

Among respondents, 54% said that they came up with the idea to start their firm during a conversation with other people, which provides strong anecdotal evidence for the importance of knowledge spillovers during face-to-face conversations. Many of these conversations happened with colleagues, which is interesting because these occur between people who had selected themselves into specific firms and thus probably have similar knowledge and expertise. Family members and friends are also high up on the list of people the entrepreneurs interacted with when coming up with their idea. It is, however, interesting to note that only 28% of the individuals started the firm with someone else, showing that even if they *came up with the idea* with someone else, the majority (72%) did not *collaborate* with another when starting the actual firm.

We first asked the respondents to recall where they came up with the idea to start a new firm (Table 6, column 1) and where they would meet up to discuss the starting up of the firm if they started with other people (column 2). It is worth noting that several alternatives could be selected, and thereby the numbers do not necessarily add up to 100%.

Starting with column 1, most of the entrepreneurs responded that they came up with the idea either at home or in the office. Only 2% claimed that they came up with the idea in a bar, café, or restaurant. Among the 'other' alternatives, the majority mentioned that they could not pinpoint one place and that it was a process involving several or all the alternatives suggested. A few of the individuals mentioned their school or university as another common place.

As for planning the starting of the firm in column 2, 24% of respondents cited third places such as restaurants and cafés in this stage. Among the 28% of individuals who started the firm with someone else, when they were asked about the place where they would meet to plan the starting up of the organisation, the most common answers were home and office. It is interesting to note that very few of the respondents mentioned the gym, conferences, nature or the online option for either coming up with the idea or during planning the start-up phase.

We next directly asked about which specific meeting places were considered important in the early stages of the start-up in an open-answer question. About 20% of those who responded mentioned third places including bars, restaurants, and cafés as very important in the start-up phase, where they would mostly meet other people to plan or discuss issues related to the firm. Moreover, events such as fairs or concerts, where they would meet people of similar interests, also showed up as important third places. In these written answers, 21% also mentioned the office as an important meeting place in the beginning, while approximately 15% discussed that most meetings they had were held online.

We further asked the entrepreneur respondents to rank on a scale from 1 to 5 (where 5 is very important) how important accessibility to physical meeting places (outside of home and work) was for them when starting up their firm. Figure 2 shows the distribution of the results.

Most of the entrepreneurs answered 4 or above on this question, which is in line with our initial regression analysis. To dig deeper into the distribution of this question, we separated the answers by one-time entrepreneurs and serial entrepreneurs who started several firms. Due to the low number of observations, we recoded the answers according to three alternatives[5] instead of five (Figure 3).

The results show that accessibility to meeting places is more important for serial entrepreneurs compared with one-timers. The difference in responses is statistically significant between the two groups. This is an interesting

Table 5. Descriptive statistics for the survey sample and the population of start-ups, 2016.

	Survey	Start-ups
Size	1.64	1.20
Share of educated entrepreneurs	0.67	0.51
Age of the entrepreneur	45	44
Serial entrepreneur	0.41	0.39
Still has the firm 2020	0.87	0.37

Table 6. Descriptive data from the telephone survey.

Place	Came up with the idea (1)	Planned the starting of the firm (2)
Office	41%	25%
Home	38%	40%
Conference	0%	2%
Third place (bar, café, restaurant)	2%	24%
Gym	0.5%	0%
Nature	2.3%	2%
Online	0.5%	3%
Other	30%	21%

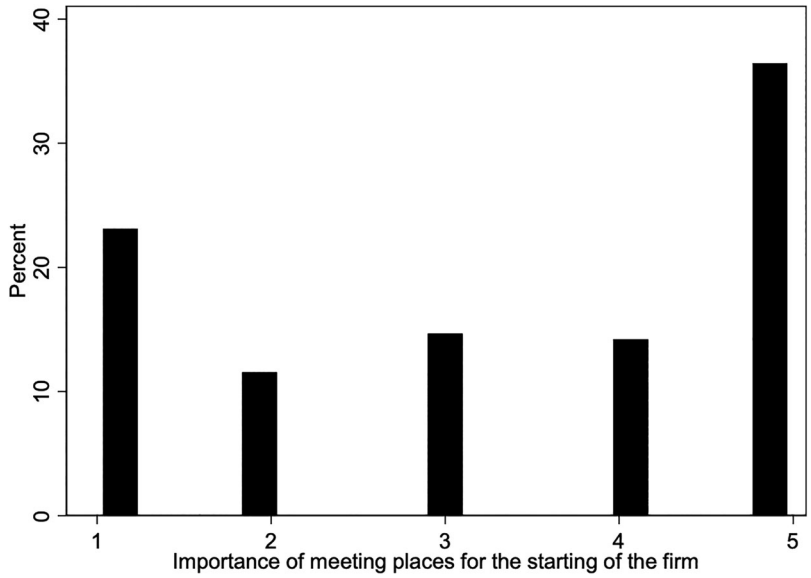


Figure 2. Importance of meeting places for the starting of the firm.

insight because it suggests that serial entrepreneurs, to a higher extent than one-time entrepreneurs, may rely more on social interactions and face-to-face meetings for the flow of ideas and knowledge exchange. This is a potential learning for entrepreneurs as it indicates that connectivity might be an important consideration in planning a start-up.

Figure 4 presents the distributions when separating between those who came up with the idea during a conversation with someone else compared with those who thought of it alone. The results suggest that the entrepreneurs who tapped into conversations also value access to meeting places more.

Overall, these results indicate that entrepreneurs are generally a heterogeneous group, but for those who started more than one firm and those who developed the idea through conversations, the value of third places is higher.

Those entrepreneurs who answered 4 or above on this question were also asked to elaborate in open answers on what they valued with third places. The open answers were broad, but one answer that was often given in this question is that physical meeting places facilitate contact and the flow of ideas and creativity. This is in line with our hypotheses as well as with our empirical findings as stated earlier in the paper. Related reasons that showed up are that it is easier and more convenient to discuss

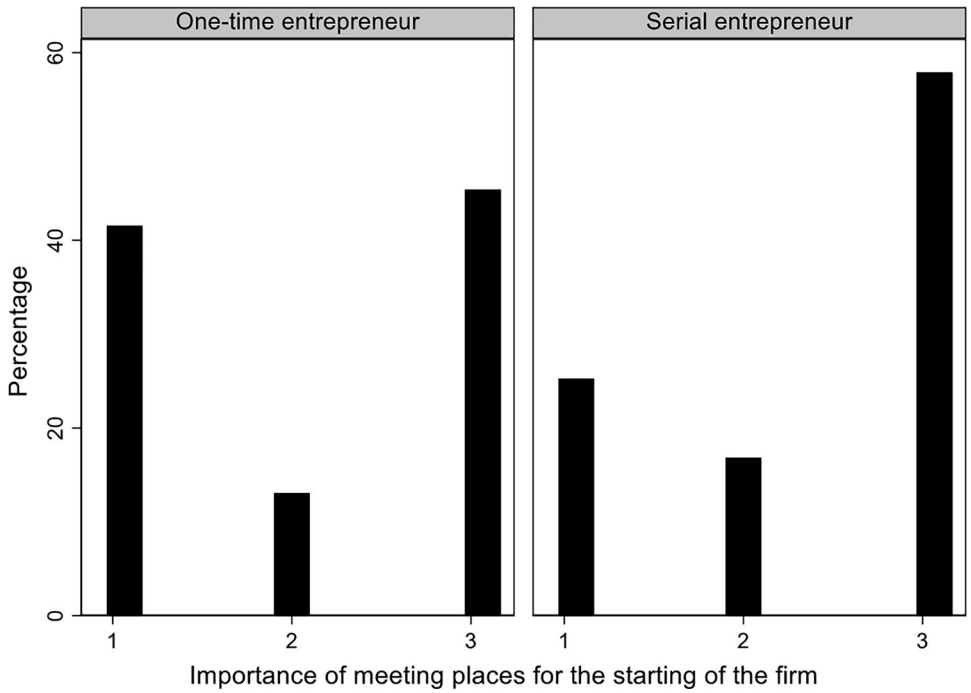


Figure 3. Importance of meeting places for the starting of the firm by type of entrepreneur.

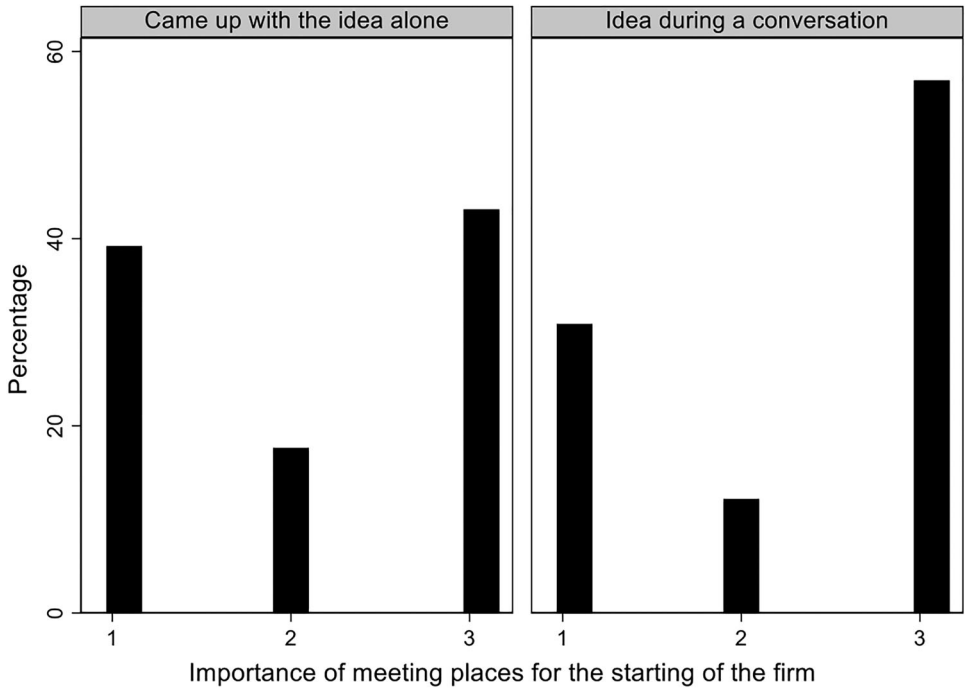


Figure 4. Importance of meeting places for the starting of the firm by collaboration type.

with other people face-to-face rather than online, and easier to build trust as well as to create better and stronger relationships with their clients.

In general, the survey results show that while access to meeting places was not crucial for the development of an idea to start a firm (which might be difficult to remember exactly how it happened), entrepreneurs in general, and serial entrepreneurs more specifically, see them as important in the start-up formation process. Somewhat counter-intuitively, we might interpret these survey results as supporting the importance of third places primarily in the formal (i.e., planning) rather than the informal (i.e., ideation) stage of the start-up process. This is interesting because the idea that places outside of the home or work could form part of the crucial infrastructure for *formal* start-up activities is currently under-theorised. Most of the existing literature assumes the entrepreneurial benefits from third places come from fostering informal interactions and that lead to knowledge spillovers (e.g., Jacobs, 1961, 1969), but these theories were largely developed in a time before it was possible to conduct much of the formal work of planning and organising a firm online using a laptop at a café or restaurant. It is certainly possible that this paper's empirical results are supporting an alternative mechanism through which third places foster entrepreneurship, that is, as 'free' quasi-public venues for formally planning and organising the firm in its early stages. This certainly deserves to be studied more in future work.

But does this mean that the firms where the meeting places were of higher importance perform better as well? We have two indirect measures of performance in the survey: whether the firm is still active in 2020 (at the point in time when we conducted the interview) and the size of the firm. The results are shown in Figures 5 and 6.

The results show that meeting places were more important for those entrepreneurs who still have an active firm in 2020 (86% of the respondents). We also see a similar pattern on the size of the firms; the ones whose firm is the largest, the importance of meeting places looks to be of higher levels. These are of course very descriptive results, and do not necessarily imply that the ones who state that meeting places are important for the start of the firm perform better. However, the results seem to support that argument.

6. DISCUSSION AND CONCLUSIONS

This paper provides a detailed empirical assessment of the role of third places in high-tech start-up formation in cities. We argue that third places constitute a critical piece of the innovation infrastructure, knowledge exchange and connectivity of cities. We use detailed geo-coded register data between 2010 and 2019 to observe the relationship between third places and start-ups over time in Stockholm. To deal with identification issues, we rely on neighbourhood fixed effects and IV estimates, and we complemented the results from the register longitudinal data with a telephone survey providing further confirmation of the findings.

Our empirical results provide suggestive evidence that third places such as cafés, bars and restaurants facilitate knowledge spillovers among people and provide a kind of supportive quasi-public start-up infrastructure for new firm founders, which in turn leads to a higher rate of start-ups. This supports the findings of other recent research in this area (Fang et al., 2022). Our primary finding is that third places play a key role in supporting innovation and entrepreneurship within cities, particularly in

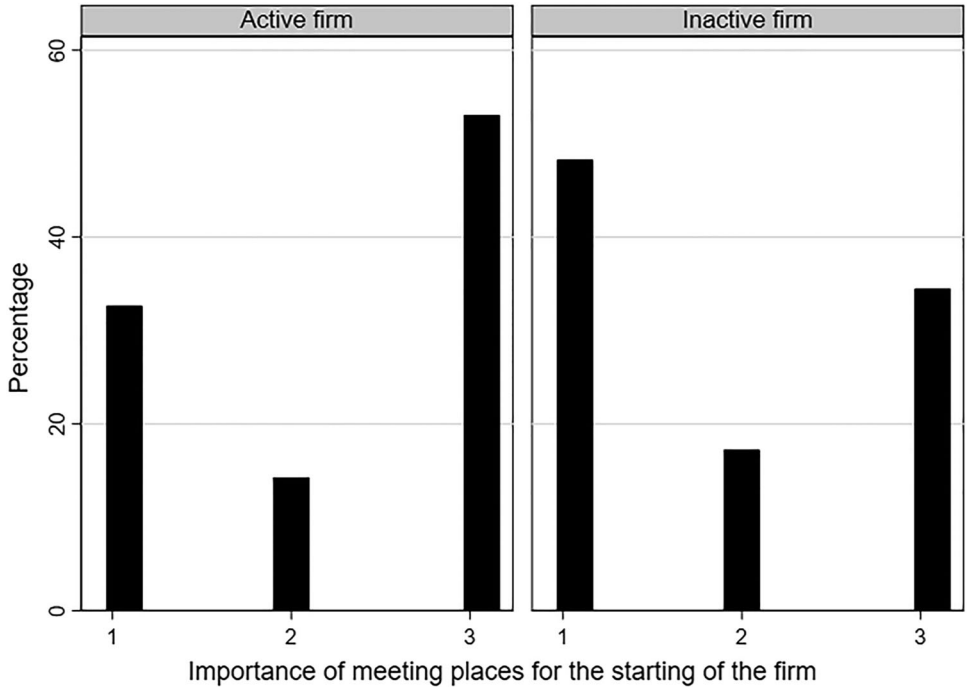


Figure 5. Importance of meeting places for starting the firm by whether the firm is still active in 2020.

the planning stages of the firm (as shown in the survey results). While this result is somewhat counterintuitive from a theoretical perspective, given the focus of the existing literature on the role of third places in fostering informal knowledge spillovers, we think that it provides an interesting opportunity to develop a stronger theory of the role of third places on start-up activities that include the relatively recent ability for entrepreneurs to conduct a vast amount of *formal* start-up planning activities online from a local coffee shop or bar. However, it is worth noting that this paper does not say anything regarding the potential substitutive role that other types of places where people can conduct business and discuss ideas in a face-to-face environment, such as libraries and community centres, might play. That is another interesting avenue for future research on this topic.

While the identification of a direct causal effect of third places on the aggregate level of entrepreneurship in this case is difficult, these findings are useful to policymakers at both the local/regional and national levels.

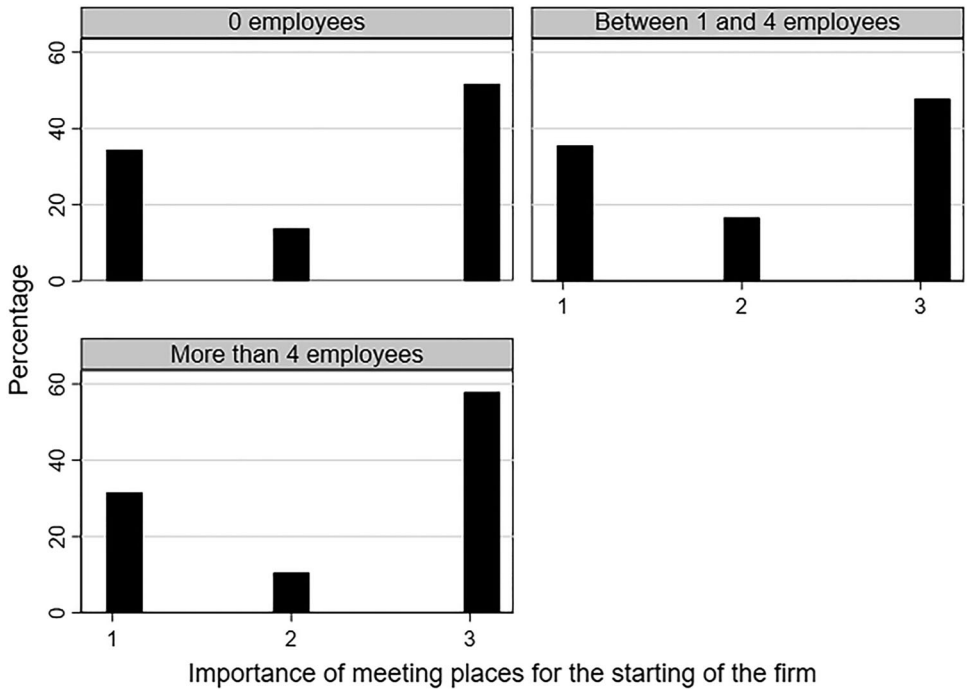


Figure 6. Importance of meeting places for starting the firm, separated by firm size.

From an urban spatial planning perspective, attracting entrepreneurship *to* particular areas – whether or not an aggregate increase is observed – is an important policy objective for a number of reasons, including increasing compact development, reducing spatial inequality and encouraging reinvestment and redevelopment. At the same time, in the context of agglomeration and knowledge spillovers, we might expect secondary (aggregate) benefits to innovation and/or entrepreneurship based purely on creating a denser spatial pattern of start-ups within the city, which could help accomplish national economic development goals related to innovation and economic growth. In that way, our results suggest that planning policies encouraging the establishment of third places in particular areas might also be used to help generate an innovative infrastructure for new high-tech business creation within cities.

Our findings may have special relevance for cities that are crafting strategies to revitalise their downtown areas that were hard hit by the COVID-19 pandemic and the shift to remote work it motivated (Barrero et al., 2021). This not only led to a reduction of office space but also impacted many of the third places, such as restaurants, cafés and bars in these areas which were closed for a while and in some cases have been slow to reopen. It may be that cities would be better off focusing on repositioning their downtowns and business areas more around third spaces and their connectivity functions as they strive to revitalise their downtown areas.

Some work (Barrero et al., 2021; Delventhal et al., 2022) suggests that the pandemic has resulted in a new way of working, where more people will work from home. This could in turn have an impact on the demand for downtowns and third places. However, a series of recent research shows how those who have the greatest ability to work from home are also individuals with a preference for high-density neighbourhoods (Brueckner et al., 2023; Veuger et al., 2023), which may be an effect of the demand for urban amenities and social interactions outside of the home. Similar results have been found in the case of Stockholm (Correa, 2023).

In any event, our research strongly suggests that third places – as joint arenas for production and consumption – play a key role in supporting innovation, start-ups and economic development in cities.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the authors.

ETHICS STATEMENT

Informed consent on the research project, data collection and anonymisation process was provided to the survey participants in this study. Ethical approval as a part of the larger project 'Geographical Aspects on Economic Performance and Individual Wellbeing' (Dnr: 2018/174-31) was provided by the Swedish Ethical Review Authority in Linköping, Sweden.

NOTES

1. For more information about RegSO, see Statistics Sweden at https://www.scb.se/hitta-statistik/regional-statistik-och-kartor/regionala-indelningar/regso—regionala-statistikomraden/
2. We naturally do not rule out the role that other third places, besides the ones we include, are also important for knowledge spillovers, for example, places such as gyms or beauty salons. However, to keep the analysis clear, we try not to have a too heterogenous definition of third places.
3. We also included accessibility to venture capitalists as a control variable, which showed to be significant and positively related to entrepreneurship. However, we only have data for 2012 and 2013 for that variable, and therefore we do not include it in the main estimations. The results are available upon request.
4. High-tech firms are defined as belonging to the high-tech, medium high-tech and knowledge-intensive services according to the Eurostat classification.
5. Alternatives 1 and 2 were recoded as 1, alternative 3 was recoded as 2, and alternatives 4 and 5 were recoded as 3.

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